Adaptive Graph-aware Reinforcement Learning for Dynamic Parameter Control in Ant Colony Optimization

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***Abstract*—Ant Colony Optimization (ACO) has been widely adopted for solving routing and combinatorial optimization problems due to its distributed search capability and robustness. However, the performance of conventional ACO heavily depends on manually configured parameters, which remain fixed throughout the optimization process and often become suboptimal in dynamic environments where graph structures and edge conditions continuously change. This limitation reduces the adaptability and convergence efficiency of ACO when addressing Dynamic Traveling Salesman Problems (DTSPs). This paper proposes an Adaptive Graph-aware Reinforcement Learning framework for dynamic parameter control in ACO. The proposed approach employs a Deep Deterministic Policy Gradient (DDPG) agent to continuously adjust pheromone-related parameters during the optimization process based on a graph-aware state representation. The state representation integrates both graph structural characteristics and optimization-related features, enabling the reinforcement learning agent to perceive environmental changes and adapt the search strategy accordingly. The learned policy dynamically balances exploration and exploitation, thereby improving the robustness of ACO under dynamic routing scenarios. Experiments are conducted on simulated dynamic routing environments with time-varying edge conditions and graph perturbations. The proposed method is compared with conventional ACO and a Q-learning-based adaptive parameter control approach using multiple evaluation metrics, including solution quality, convergence speed, and adaptability to environmental changes. Experimental results demonstrate that the proposed framework consistently achieves superior optimization performance while maintaining stable convergence under dynamic conditions. The proposed graph-aware adaptive parameter control strategy provides an effective integration of reinforcement learning and swarm intelligence, offering a promising solution for intelligent routing and dynamic combinatorial optimization problems.**

***Keywords*-Dynamic Traveling Salesman Problem, Ant Colony Optimization, Deep Reinforcement Learning, Graph-aware Learning, Adaptive Parameter Control, Route Optimization.**

I. Introduction

Many real-world optimization problems, including intelligent transportation systems, logistics planning, autonomous vehicle routing, and delivery services, can be formulated as graph-based combinatorial optimization problems. Among them, the Traveling Salesman Problem (TSP) has long served as a fundamental benchmark due to its NP-hard complexity and broad applicability. In practical scenarios, however, transportation networks are inherently dynamic, where edge costs may vary over time because of traffic congestion, road closures, accidents, or changing environmental conditions. These characteristics give rise to the Dynamic Traveling Salesman Problem (DTSP), in which routing decisions must continuously adapt to evolving graph conditions while maintaining high solution quality. Compared with the static TSP, DTSP introduces significantly greater challenges because optimization algorithms must rapidly respond to environmental changes instead of searching for a single static optimum [1], [2].

Ant Colony Optimization (ACO), originally inspired by the foraging behavior of ant colonies, has become one of the most successful swarm intelligence algorithms for solving routing and combinatorial optimization problems. Since its introduction, numerous variants, including the Ant System (AS), Ant Colony System (ACS), and Max-Min Ant System (MMAS), have demonstrated competitive performance across a wide range of routing and graph optimization tasks [3], [4], [2]. Owing to its distributed search mechanism and positive feedback through pheromone updating, ACO is particularly suitable for solving graph-based optimization problems. Nevertheless, the performance of conventional ACO is highly sensitive to several algorithmic parameters, such as the pheromone

influence coefficient, evaporation rate, and heuristic weighting factor. These parameters are typically determined empirically before optimization and remain fixed throughout the search process. Such a static parameter configuration often leads to premature convergence, insufficient exploration, and degraded optimization performance when the underlying graph changes dynamically [5], [2].

To overcome the limitations of manually tuned parameters, adaptive parameter control has become an active research direction in swarm intelligence and evolutionary optimization. Existing approaches mainly employ heuristic rules, fuzzy logic, self-adaptive mechanisms, or meta-optimization strategies to dynamically adjust algorithm parameters during the search process. More recently, Reinforcement Learning (RL) has emerged as an effective paradigm for sequential decision-making in uncertain and dynamic environments. Through continuous interaction with the optimization environment, an RL agent can learn adaptive control policies that automatically balance exploration and exploitation without relying on handcrafted parameter schedules. The rapid development of Deep Reinforcement Learning (DRL), particularly actor–critic algorithms such as Deep Deterministic Policy Gradient (DDPG), has further demonstrated remarkable capability in solving continuous control problems and optimization tasks with high-dimensional state spaces [6], [7].

Recent studies have shown the potential of integrating machine learning with combinatorial optimization to improve the adaptability of classical optimization algorithms. Learning-based optimization methods can exploit historical search experience to guide future decisions, thereby reducing manual intervention and improving generalization across problem instances [8], [9]. Nevertheless, most existing RL-assisted ACO methods primarily adjust algorithm parameters according to optimization statistics while paying limited attention to the structural characteristics of the underlying graph. In dynamic routing environments, graph topology, connectivity, and edge availability provide valuable contextual information that directly influences search behavior. Ignoring these structural properties limits the ability of the learning agent to accurately perceive environmental changes and select appropriate parameter adjustment strategies. Furthermore, many existing adaptive optimization methods are evaluated primarily on static benchmark problems, making their effectiveness in highly dynamic routing environments insufficiently investigated [9].

To address these limitations, this paper proposes an **Adaptive Graph-aware Reinforcement Learning** framework for dynamic parameter control in Ant Colony Optimization. The proposed framework integrates a Deep Deterministic Policy Gradient (DDPG) agent with a graph-aware state representation that combines graph structural characteristics and optimization-related information. Based on the observed environment, the RL agent dynamically adjusts pheromone-related parameters throughout the optimization process, enabling the search strategy to continuously adapt to changing graph conditions. By jointly exploiting graph information and reinforcement learning, the proposed framework aims to improve convergence efficiency, solution quality, and robustness in dynamic routing environments.

The main contributions of this paper are summarized as follows:

- We propose an adaptive reinforcement learning framework that dynamically controls the key parameters of Ant Colony Optimization instead of relying on manually predefined parameter settings.
- We design a graph-aware state representation that integrates graph structural characteristics with optimization-related features, enabling the reinforcement learning agent to better perceive environmental dynamics.
- We develop a DDPG-based adaptive parameter controller that continuously adjusts pheromone-related parameters during the optimization process, thereby achieving a more effective balance between exploration and exploitation.
- We conduct comprehensive experiments on dynamic routing scenarios and compare the proposed method with conventional ACO and a Q-learning-based adaptive parameter control approach. Experimental results demonstrate that the proposed framework consistently improves convergence speed, solution quality, and adaptability under dynamic graph environments.

The remainder of this paper is organized as follows. Section II reviews related work on Ant Colony Optimization, dynamic routing problems, reinforcement learning, and learning-based combi-

natorial optimization. Section III presents the proposed Adaptive Graph-aware Reinforcement Learning framework. Section IV describes the experimental settings and discusses the comparative results. Finally, Section V concludes the paper and outlines future research directions.

II. Related Work

A. Ant Colony Optimization for Routing Problems

Ant Colony Optimization (ACO) is one of the most representative swarm intelligence algorithms for solving graph-based combinatorial optimization problems. Since the introduction of the Ant System (AS) by Dorigo *et al.* [3], numerous variants have been proposed to improve convergence speed and solution quality. The Ant Colony System (ACS) introduces local pheromone updating and a pseudorandom proportional rule to enhance the balance between exploration and exploitation [4]. Subsequently, the Max–Min Ant System (MMAS) further improves search stability by restricting pheromone values within predefined bounds, thereby reducing the risk of premature convergence [2]. Owing to these advantages, ACO has been successfully applied to a wide range of routing, scheduling, and network optimization problems.

For dynamic routing problems, conventional ACO exhibits an inherent advantage due to its distributed search mechanism and indirect communication through pheromone trails. Nevertheless, when graph structures or edge costs change dynamically, pheromone information accumulated during previous iterations may become outdated, resulting in slower adaptation and degraded optimization performance. Existing adaptive ACO variants generally employ heuristic rules or manually designed parameter adjustment strategies to improve adaptability; however, these methods often require extensive parameter tuning and exhibit limited generalization across different optimization scenarios [5], [2].

B. Reinforcement Learning for Combinatorial Optimization

Recent advances in Reinforcement Learning (RL) have demonstrated considerable potential for solving combinatorial optimization problems. Unlike conventional optimization algorithms that rely on handcrafted heuristics, RL learns decision policies directly through interactions with the environment, enabling adaptive optimization under uncertain and dynamic conditions [7]. The emergence of Deep Reinforcement Learning (DRL) has significantly expanded the capability of RL by combining reinforcement learning with deep neural networks to approximate complex value functions or policies. In particular, the Deep Deterministic Policy Gradient (DDPG) algorithm extends actor–critic learning to continuous action spaces, making it well suited for continuous parameter optimization problems [6].

Beyond classical reinforcement learning, machine learning techniques have increasingly been integrated into combinatorial optimization. Khalil *et al.* proposed a graph embedding framework for learning optimization algorithms directly from graph instances, demonstrating that learned policies can outperform handcrafted heuristics on several graph optimization tasks [8]. Bengio *et al.* further surveyed recent developments in machine learning for combinatorial optimization and highlighted learning-based optimization as a promising direction for improving both solution quality and computational efficiency [9]. These studies suggest that reinforcement learning provides an effective paradigm for enhancing traditional optimization algorithms through adaptive decision-making.

C. Learning-based Parameter Control in Metaheuristics

Parameter control has long been recognized as a critical factor influencing the performance of metaheuristic algorithms. Conventional approaches generally determine parameter values empirically or through offline tuning before optimization begins. However, fixed parameter settings are often insufficient for dynamic optimization problems because the search process continuously evolves and requires different exploration strategies at different stages.

To address this issue, adaptive parameter control strategies have been developed using heuristic rules, self-adaptive mechanisms, evolutionary strategies, and reinforcement learning. Compared with heuristic-based approaches, reinforcement learning offers the advantage of learning parameter adjustment policies directly from optimization feedback without requiring manually designed adaptation rules. Recent research has demonstrated that RL-based parameter control can improve convergence behavior and robustness in evolutionary algorithms and swarm intelligence methods, particularly in dynamic optimization environments [7], [9]. Nevertheless, most existing studies focus

primarily on optimization performance indicators, such as fitness improvement or convergence rate, while relatively little attention has been paid to exploiting graph structural information during the parameter adaptation process.

D. Research Gap and Motivation

The literature review reveals two important research gaps. First, although reinforcement learning has shown promising results for adaptive optimization, most existing RL-assisted ACO methods represent the optimization state using only algorithm-specific statistics, such as pheromone distribution, fitness improvement, or convergence behavior. Consequently, the learning agent has limited awareness of the structural changes occurring in the underlying graph. Second, the majority of existing adaptive parameter control methods have been evaluated primarily on static benchmark problems, while their effectiveness in dynamic routing environments remains insufficiently explored.

Motivated by these limitations, this paper proposes an Adaptive Graph-aware Reinforcement Learning framework for dynamic parameter control in Ant Colony Optimization. Unlike conventional adaptive approaches, the proposed framework incorporates both graph structural characteristics and optimization-related features into the state representation, enabling the reinforcement learning agent to perceive environmental changes more effectively and dynamically adjust pheromone-related parameters throughout the optimization process.

III. Proposed Model

IV. Experimental Results

V. Conclusion

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